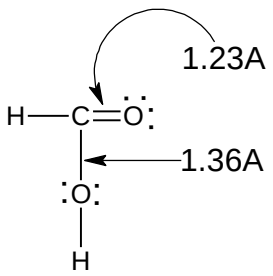


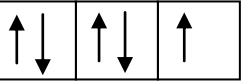
NO	PART	SCHEME
1	(a)	$2\text{NO(g)} + 2\text{CO(g)} \rightarrow \text{N}_2\text{(g)} + 2\text{CO}_2\text{(g)}$ $2 \text{ mol NO} \equiv 2 \text{ mol CO} \equiv 1 \text{ mol N}_2 \equiv 2 \text{ mol CO}_2$ $\text{Mol CO} = \frac{\text{mass of compound}}{\text{molar mass of compound}}$ $= \frac{300\text{g}}{28 \text{ g mol}^{-1}} @$ $= 10.71\text{g}$ $\text{Mass N}_2 = \frac{10.71}{2} \times 28 \text{ g mol}^{-1} @$ $= 150.08\text{g}$ $\text{Mass CO}_2 = 10.71 \times 44 \text{ g mol}^{-1} @$ $= 471.24\text{g}$ $\% \text{CO}_2 = \frac{471.24}{471.24 + 150.08} \times 100$ $= 75.8\%$
	(b)(i)	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ $(\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-) \times 6$ $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Fe}^{2+} \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$
	(b)(ii)	$\text{Mol Na}_2\text{Cr}_2\text{O}_7 = 50.0 \times 10^{-3} \text{ L} \times 0.025 \text{ mol L}^{-1}$ $= 1.25 \times 10^{-3} \text{ mol}$ $\text{Mass Na}_2\text{Cr}_2\text{O}_7 = 1.25 \times 10^{-3} \text{ mol} \times 262.0 \text{ g mol}^{-1}$ $= 0.33\text{g}$
	(b)(iii)	$\text{Mol Cr}_2\text{O}_7^{2-} = 30.00 \times 10^{-3} \times 0.025$ $= 7.5 \times 10^{-4} \text{ mol}$ $1 \text{ mol Cr}_2\text{O}_7^{2-} \equiv 6 \text{ mol Fe}^{2+}$ $7.5 \times 10^{-4} \text{ mol Cr}_2\text{O}_7^{2-} \equiv 7.5 \times 10^{-4} \times 6 \text{ mol} @ 4.5 \times 10^{-3} \text{ mol Fe}^{2+}$ $\text{Molarity} = \frac{\text{no of moles of solute}}{\text{volume of solution in L}}$ $= \frac{4.5 \times 10^{-3}}{40 \times 10^{-3}} = 0.112\text{M}$

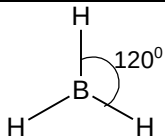
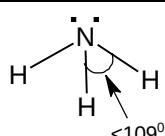
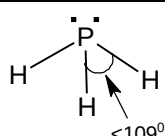
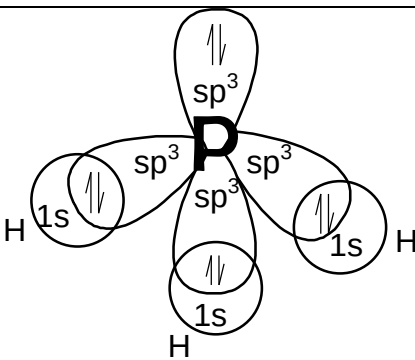
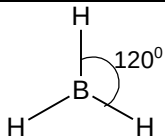
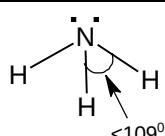
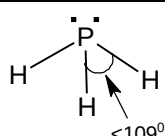
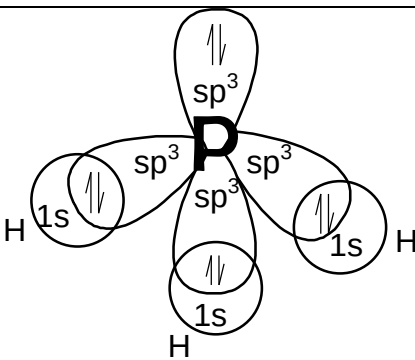
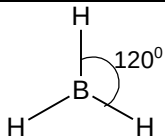
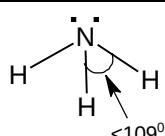
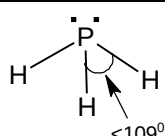
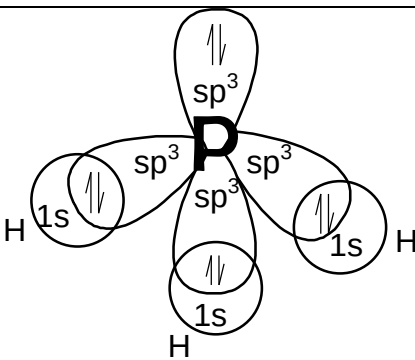
NO	PART	SCHEME
2	(a)	 Double bond is shorter than single bond MUST SHOW THE LEWIS STRUCTURE MUST SHOW THE BOND LENGTH
	(b)(i)	XeF_4 Total no of electrons = $8 + (1 \times 4) = 12$ No of electron pairs = $12/2 = 6$ Thus, no of bonding pairs = 4, lone pair = 2 XeO_2F_2 Total no of electrons = $8 + (2 \times 2) + (1 \times 2) = 14$ No of electron pairs = $14/2 = 7$ Thus, no of bonding pairs = 4@5@6, lone pair = 1
	(b)(ii)	XeF_4 = square planar XeO_2F_2 = distorted tetrahedron @ see-saw
	(b)(iii)	$\text{XeF}_4 = \text{sp}^3\text{d}^2$ $\text{XeO}_2\text{F}_2 = \text{sp}^3\text{d}$
	(b)(iv)	XeF_4 = non-polar XeO_2F_2 = polar

NO	PART	SCHEME
3	(a)(i)	Since $\text{CO}_2(\text{g})$ in the reaction is collected over water, $P_{\text{total}} = P_{\text{H}_2\text{O}} + P_{\text{CO}_2}$ $P_{\text{H}_2\text{O}} = 31.8 \text{ mm Hg} \times \frac{1 \text{ atm}}{760 \text{ mm Hg}} @ 4.18 \times 10^{-2} \text{ atm}$ $P_{\text{CO}_2} = P_{\text{total}} - P_{\text{H}_2\text{O}}$ $= 1 \text{ atm} - 4.18 \times 10^{-2} \text{ atm}$ $= 0.958 \text{ atm}$ $n_{\text{CO}_2} = \frac{P_{\text{CO}_2} V}{RT}$ $= \frac{(0.958 \text{ atm})(0.1073 \text{ L})}{(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(303.15 \text{ K})}$ $= \mathbf{4.13 \times 10^{-3} \text{ mol}}$
	(a)(ii)	$1 \text{ mol CO}_2 \equiv 1 \text{ mol CaCO}_3$ $n_{\text{CaCO}_3} = n_{\text{CO}_2} = 4.13 \times 10^{-3} \text{ mol}$ $n_{\text{CaCO}_3} = n_{\text{CO}_2} \times M_w$ $= 4.13 \times 10^{-3} \text{ mol} \times (40.1 + 12.0 + 3(16.0)) \text{ g mol}^{-1}$ $= \mathbf{0.414 \text{ g}}$
	(b)	(Choose any three) Ionic Solid: ionic bonds Molecular solid: Van der Waals forces Network solid @ Giant covalent: Covalently bonded atoms Metallic solid: metallic bonds @ metal cations with a cloud of delocalized electrons

NO	PART	SCHEME												
4	(a)(i)	$\text{HA(aq)} \rightleftharpoons \text{H}^{\text{+}}(\text{aq}) + \text{A}^{\text{-}}(\text{aq})$<table><tr><td>Initial</td><td>1.12</td><td>0</td><td>0</td></tr><tr><td>Δ</td><td>-x</td><td>+x</td><td>+x</td></tr><tr><td>$\rightleftharpoons$</td><td>1.12 -x</td><td>x</td><td>x</td></tr></table>$K_{\text{a}} = \frac{[\text{H}^{\text{+}}][\text{A}^{\text{-}}]}{[\text{HA}]}$$2.95 \times 10^{-8} = \frac{[\text{x}][\text{x}]}{[1.12 - \text{x}]}$ Since K_{a} is very very small, so x in denominator is negligible @ $1.12 - x \approx 1.12$ $[\text{H}^{\text{+}}] = x = \sqrt{(2.95 \times 10^{-8})(1.12)}$$= 1.82 \times 10^{-4}$ pH = -log $[\text{H}^{\text{+}}]$ = - log (1.82×10^{-4}) = 3.74	Initial	1.12	0	0	Δ	-x	+x	+x	$\rightleftharpoons$	1.12 -x	x	x
Initial	1.12	0	0											
Δ	-x	+x	+x											
$\rightleftharpoons$	1.12 -x	x	x											
	(a)(i)	$\% \text{ dissociation} = \frac{1.82 \times 10^{-4}}{1.12} \times 100$$= \mathbf{1.62 \times 10^{-2}}$												
	(b)(i)	$\text{Ca}_3(\text{PO}_4)_2(\text{s}) \rightleftharpoons 3\text{Ca}^{2+}(\text{aq}) + 2\text{PO}_4^{3-}(\text{aq})$$\mathbf{K_{sp} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2}$												
	(b)(ii)	$\text{Ca}_3(\text{PO}_4)_2(\text{s}) \rightleftharpoons 3\text{Ca}^{2+}(\text{aq}) + 2\text{PO}_4^{3-}(\text{aq})$<table><tr><td></td><td>3x</td><td>2x</td></tr></table>$K_{\text{sp}} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2$$1.2 \times 10^{-26} = (3x)^3 (2x)^2$$@ 1.2 \times 10^{-26} = 108x^5$$x = 2.56 \times 10^{-6} \text{M (molar solubility of Ca}_3(\text{PO}_4)_2)$ $[\text{Ca}^{2+}] = 3x$$= 3 \times 2.56 \times 10^{-6}$$= \mathbf{7.68 \times 10^{-6} \text{ M}}$		3x	2x									
	3x	2x												

		$[\text{PO}_4^{3-}] = 2x$ $= 2 \times 2.56 \times 10^{-6}$ $= 5.12 \times 10^{-6} \text{ M}$
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NO	PART	SCHEME
5		Atom X : $1s^2 2s^2 2p^6 3s^2 3p^5$ @  1s  2s  2p  3s  3p 1. Aufbau Principle • Electrons will fill orbitals in the increasing order of orbital energy level • The increasing order of energy are as $1s < 2s < 2p < 3s < 3p$ 2. Hund's Rule • Most stable arrangement of electrons in orbitals of a subshell is the one with the greatest number of parallel spins • Thus, electrons fill each and all degenerate orbitals single before they pair up as in the 2p and 3p orbitals 3. Pauli's Exclusion Principle • No two electrons in an atom can have the same set of four quantum numbers • Only two electrons may occupy the same orbital, thus these electrons must have opposite spins
		The extra electron is added to the same shell ($n=3$) Extra repulsion is produced by the incoming electron causing the atom to expand As a result, X^- ion is bigger than atom X
		X, Y and Z are located in Period 3, 4 and 5 respectively. As we go down the group from X to Z, new shell (n increases) and inner electrons are being added thus shielding effect increases Furthermore, valence electrons being far away from the nucleus so the attractive forces of electrons become smaller Therefore the electronegativity of atoms will decrease.

NO	PART	SCHEME																										
6		VSEPR: The repulsion of bond pair-bond pair < bond pair-lone pair < lone pair- lone pair <table><tr><th>BH₃</th><th>NH₃</th><th>PH₃</th></tr><tr><td></td><td colspan="2">N and P in the same group, same number of valence electrons</td></tr><tr><td>3 bonding pairs</td><td colspan="2">3 bonding pairs, 1 lone pair</td></tr><tr><td>Trigonal planar</td><td colspan="2">Trigonal pyramidal</td></tr><tr><td>120</td><td colspan="2">Bond angle: <109.5</td></tr><tr><td></td><td></td><td></td></tr><tr><td>Bonding pair-bonding pair repulsion</td><td colspan="2">Lone pair-bonding pair repulsion > bonding pair-bonding pair repulsion</td></tr></table> <table><tr><td>Lone pair (sp³)</td><td rowspan="4"></td></tr><tr><td>Bond pair (sp³ – s)</td></tr><tr><td>Shape of sp³ orbital</td></tr><tr><td>Shape of s orbital</td></tr></table>	BH ₃	NH ₃	PH ₃		N and P in the same group, same number of valence electrons		3 bonding pairs	3 bonding pairs, 1 lone pair		Trigonal planar	Trigonal pyramidal		120	Bond angle: <109.5					Bonding pair-bonding pair repulsion	Lone pair-bonding pair repulsion > bonding pair-bonding pair repulsion		Lone pair (sp ³)		Bond pair (sp ³ – s)	Shape of sp ³ orbital	Shape of s orbital
BH ₃	NH ₃	PH ₃																										
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Lone pair (sp ³)																												
Bond pair (sp ³ – s)																												
Shape of sp ³ orbital																												
Shape of s orbital																												
		PH₃ is a polar molecule, presence of dipole-dipole interaction BH₃ is non-polar molecule, presence of London forces Dipole-dipole interaction > London forces @ PH₃ has stronger intermolecular forces NH₃ has hydrogen bonding NH₃ has stronger intermolecular forces than PH₃																										

NO	PART	SCHEME																								
7	(a)	Melting point increases in the order of attraction between molecules or atoms Hydrogen bonding in CH₃OH causes it to melts at a higher temperature than C₂H₆ (London forces) Since NaCl has ionic bond @ electrostatic attraction forces, it melts at a higher temperature than CH₃OH and C₂H₆ Si has giant covalent network @ giant covalent structure, highest melting point.																								
	(b)	PCl₃(g) + Cl₂(g) ⇌ PCl₅(g) <table><tr><th></th><th>PCl₃</th><th>Cl₂</th><th>PCl₅</th></tr><tr><td>Initial mol</td><td>0.0681</td><td>0.2056</td><td>0</td></tr><tr><td>Δ</td><td>-x</td><td>-x</td><td>+x</td></tr><tr><td>⇌</td><td>0.0681 - x</td><td>0.2056 - x</td><td>0.0316</td></tr><tr><td>mol</td><td>0.0365</td><td>0.1740</td><td>0.0316</td></tr><tr><td>[]</td><td>0.00365</td><td>0.01740</td><td>0.00316</td></tr></table> $K_c = \frac{[PCl_5]}{[PCl_3][Cl_2]}$$= \frac{[0.00316]}{[0.00365][0.01740]}$$= 49.76$ Δn = -1 $K_p = K_c (RT)^{\Delta n}$$= 49.76 (0.0821 \times 523)^{-1}$$= 1.16$ <u>ALTERNATIVE:</u> $P_{PCl_3} = \frac{nRT}{V} = \frac{(0.0365)(0.08206)(523.15)}{10} = 0.1567\text{atm}$$P_{Cl_2} = \frac{nRT}{V} = \frac{(0.1740)(0.08206)(523.15)}{10} = 0.7470\text{atm}$		PCl ₃	Cl ₂	PCl ₅	Initial mol	0.0681	0.2056	0	Δ	-x	-x	+x	⇌	0.0681 - x	0.2056 - x	0.0316	mol	0.0365	0.1740	0.0316	[]	0.00365	0.01740	0.00316
	PCl ₃	Cl ₂	PCl ₅																							
Initial mol	0.0681	0.2056	0																							
Δ	-x	-x	+x																							
⇌	0.0681 - x	0.2056 - x	0.0316																							
mol	0.0365	0.1740	0.0316																							
[]	0.00365	0.01740	0.00316																							

	$P_{\text{PCl}_5} = \frac{nRT}{V} = \frac{(0.0316)(0.08206)(523.15)}{10} = 0.1356\text{atm}$ $K_p = \frac{(P_{\text{PCl}_5})}{(P_{\text{PCl}_3})(P_{\text{Cl}_2})}$ $= \frac{(0.1357)}{(0.1567)(0.7470)}$ $= \mathbf{1.159}$ Le Chatelier: When a system at equilibrium is subjected to change, the system will adjust to counteract the effect Increase pressure: reaction moves forward because the number of moles of reactants is more than that of the product Increase temperature: because forward reaction is endothermic Add reactants: because the amount of reactants is more than products Remove product: because amount of product is less than reactant
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NO	PART	SCHEME
8	(a)	$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ $\text{Mol NaOH} = \frac{0.1000}{40 \text{ g mol}^{-1}} = 2.5 \times 10^{-4} \text{ mol}$ $[\text{NaOH}] = \frac{2.5 \times 10^{-4}}{\left(\frac{25}{1000}\right)} = \mathbf{0.1 \text{ M}}$ <u>Initial pH in the flask:</u> $\text{pOH} = -\log [\text{OH}^-]$ $= -\log (0.1)$ $= 1$ $\text{pH} = 14 - 1 = \mathbf{13}$

	<u>At Equivalence Point:</u> Mol of NaOH = mol of HCl $2.5 \times 10^{-4} \text{ mol of NaOH} = 2.5 \times 10^{-4} \text{ mol of HCl}$ Mol HCl = $\frac{MV}{1000}$ Volume of HCl = $\frac{(2.5 \times 10^{-4})(1000)}{0.1} = 25 \text{ ml}$ At equivalence point, only H₂O present @ neutral solution @ [H₃O⁺]=10⁻⁷ M So, pH = 7 <u>At 30ml HCl</u> NaOH is the limiting reactant Mol HCl = $\frac{(0.100)(30)}{1000} = 3 \times 10^{-3} \text{ mol}$ Excess mol HCl = $(3.0 \times 10^{-3} - 2.5 \times 10^{-3}) \text{ mol}$ = $0.5 \times 10^{-3} \text{ mol}$ Molarity of HCl = $\frac{0.5 \times 10^{-3}}{55 \times 10^{-3}} = 9.09 \times 10^{-3} \text{ M}$ pH = - log [H⁺] = - log (9.09 × 10⁻³) = 2 <u>GRAPH PLOTTED ON THE NEXT PAGE</u> Must label axis, shape and label for both strong base- strong acid titration and also for strong base- weak acid titration.
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